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MILITARY SPECIFICATION

RACK, BOMB; MULTIPLE EJECTOR, A/A37B-6 SERIES AND TRIPLE EJECTOR, A/A37B-5 SERIES

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and inspection of two types of bomb ejector racks. The Type I bomb rack is capable of carrying and releasing Up to three stores with a total combined weight not to exceed 1,360 kilograms (3,000 pounds). The Type II bomb rack is capable of carrying and releasing up to six stores with a total combined weight not to exceed 2,040 kilograms (4,500 pounds). All carried stores shall have suspension lugs spaced 355.6 millimeters (14 inches) apart and shall weigh from 22.7 kilograms (50 pounds) to 660 kilograms (1,450 pounds). The test limits in this specification define minimum acceptable capabilities.

1.2 Classification. Bomb ejector racks, hereinafter called bomb racks, covered by this specification shall be of the following types:

Type I	-A/A37B-5 Series Triple Ejector Rack (TER)
Type II	-A/A37B-6 Series Multiple Ejector Rack (MER)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93) Naval Air Engineering Center, Lakehurst, N.J. 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1095

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SPECIFICATIONS

MILITARY

MIL-I-6866	-Inspection, Penetrant Method of
MIL-T-7743	-Testing, Store Suspension and Release Equipment, General Specification for
MIL-C-81842	-Connector Assemblies for Bomb Rack Electric Fuzing Provisions

STANDARDS

MILITARY

MIL-STD-100	-Engineering Drawing Practices
MIL-STD-129	-Marking for Shipment and Storage
MIL-STD-130	-Identification Marking of U.S. Military Property
MIL-STD-143	-Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-704	-Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-794	-Parts and Equipment, Procedure for Packaging and Packing of
MIL-STD-831	-Test Reports, Preparation of
MS3314	-Lug, Suspension, 450 kg. (1,000 lb.) class, Airborne Equipment

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

DL1025AS100	-Data List, Test Fixture, MER/TER
292AS101	-Hanger, Ejector Assembly

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Order of precedence of specifications and standards. Should a conflict exist between any requirement in this specification and any requirement in the applicable documents referenced herein, the requirements of this specification shall take precedence over the referenced (applicable) document.

3. REQUIREMENTS

3.1 First article. The bomb racks furnished under this specification shall be a product which has been inspected and passed the first article inspection specified herein (see 4.3).

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Interchangeability. All parts having the same part number shall be directly and completely interchangeable with each other with respect to installation and performance.

3.4 Identification of parts. All parts shall be marked as specified on the drawings listed in the data list for the bomb rack being procured, or in accordance with MIL-STD-130, if not specified on the drawings. The item identification and part number requirements of MIL-STD-100 shall govern the part number and changes thereto. .

3.5 Materials. Materials shall be as specified on the drawings listed in the data list for the bomb rack being procured (see 6.2.1f).

3.6 Bomb rack.

3.6.1 Construction. The construction of the bomb rack shall be in accordance with the drawings listed in the data list for the bomb rack being procured and as specified herein (see 6.2.1f).

3.6.2 Performance. The bomb racks shall satisfy all performance requirements when subjected to the following inspections in the order specified in Tables I and II.

a. Bomb Rack Inspection (see 4.6.1)

b. Functional Test (see 4.6.2)

c. Arming Mechanism (see 4.6.3)

d. Electric Fuze Arming (see 4.6.4)

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e. Life (see 4.6.5)

f. Electrical Characteristics (see 4.6.6a)

g. Lag Time or Time of Release Determination (see 4.6.6b)

h. Icing (see 4.6.6c)

i. Temperature Shock (see 4.6.6d)

j. Altitude (see 4.6.6e)

k. Humidity (see 4.6.6f)

l. Sand (see 4.6.6g)

m. Salt Spray (see 4.6.6h)

n. Shock (see 4.6.6i)

o. Vibration (see 4.6.6j)

p. Lock Shut Firing (see 4.6.6k)

q. Static (see 4.6.61)

3.7 Workmanship. The workmanship displayed in fabrication and assembly of the bomb racks shall be such as to assure, within design limitations, the ability of the bomb racks to meet their performance requirements under all applicable environmental conditions specified herein. Unauthorized repair, welding, heavy burrs, or parts assembled by introduction of high stresses not prescribed in the drawings, are typical signs of inferior workmanship and shall be cause for rejection. The standards of workmanship exhibited in the approved first article sample, subject to any qualification stated in the government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure that the supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The inspection of the bomb racks shall be classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article inspection. First article inspection shall consist of all of the inspections listed in Table I. First article inspection report format shall be in accordance with MIL-STD-831 (see 6.3).

4.3.1 Sampling for first article inspection. First article bomb rack samples shall consist of the first two bomb racks manufactured on contract. The bomb racks shall be permanently labeled A and B for ease of identification. Failure of the bomb racks to pass any of the first article inspections shall be cause for rejection (see 6.2.2.1).

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the following inspections:

a. Individual Inspections. (see 4.4.1)

b. Sampling Inspections (see.4.4.2)

Quality conformance inspection report format shall be in accordance with MIL-STD-831 (see 6.3).

4.4.1 individual inspections. Individual inspections are those inspections conducted on each bomb rack. Individual inspections for the bomb racks are shown in Table II, Group 1. Failure to pass any of these inspections shall be cause for rejection of the bomb rack.

4.4.2 Sampling inspections. A random sampling of the bomb racks shall be selected by the procuring activity from each production lot in accordance with Table III and shall be subjected to the inspections listed in Table II, Group 1 and Group 2. Group 1 inspections shall be conducted prior to Group 2 inspections. Failure to pass any of these inspections shall be cause for rejection of the entire lot (see 6.2.1d).

4.5 Inspection conditions. Unless otherwise specified, all inspections shall be performed under the following conditions:

a. Room temperature. Room ambient of 25±10°C (77±18°F) indicated.

b. Test temperature. All parts of the test item shall be stabilized at the specified temperature ±2.8°C

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(±5°F) indicated prior to conducting any tests. Unless otherwise specified, temperature stabilization will have been attained when the indicated temperature of the-surface of the largest mass of the test item does not change by more than ±2.8°C (±5°F) in a period of 1 hour.

c. Swaybraces. Whenever tests require that a store be mounted to the bomb rack, swaybraces shall be used.

The swaybraces shall be adjusted as described in 4.6.2.2a.

d. Test stores. The stores, as referenced herein by weight, shall conform to the following parameters. The store suspension lugs shall be in accordance with MS3314.

1. The 22.7 kilogram (50 pound) store shall utilize applicable suspension lugs spaced 355.6 millimeters (14 inches) apart. Its center of gravity shall be midway between the suspension lugs and 51 to 152 millimeters (2 to 6 inches) below the hook bearing surface of the lug.

2. The 113 kilogram (250 pound) store shall utilize applicable suspension lugs, spaced 355.6 millimeters (14 inches) apart. Its center of gravity shall be based on the MK 81 L.D.G.P. bomb.

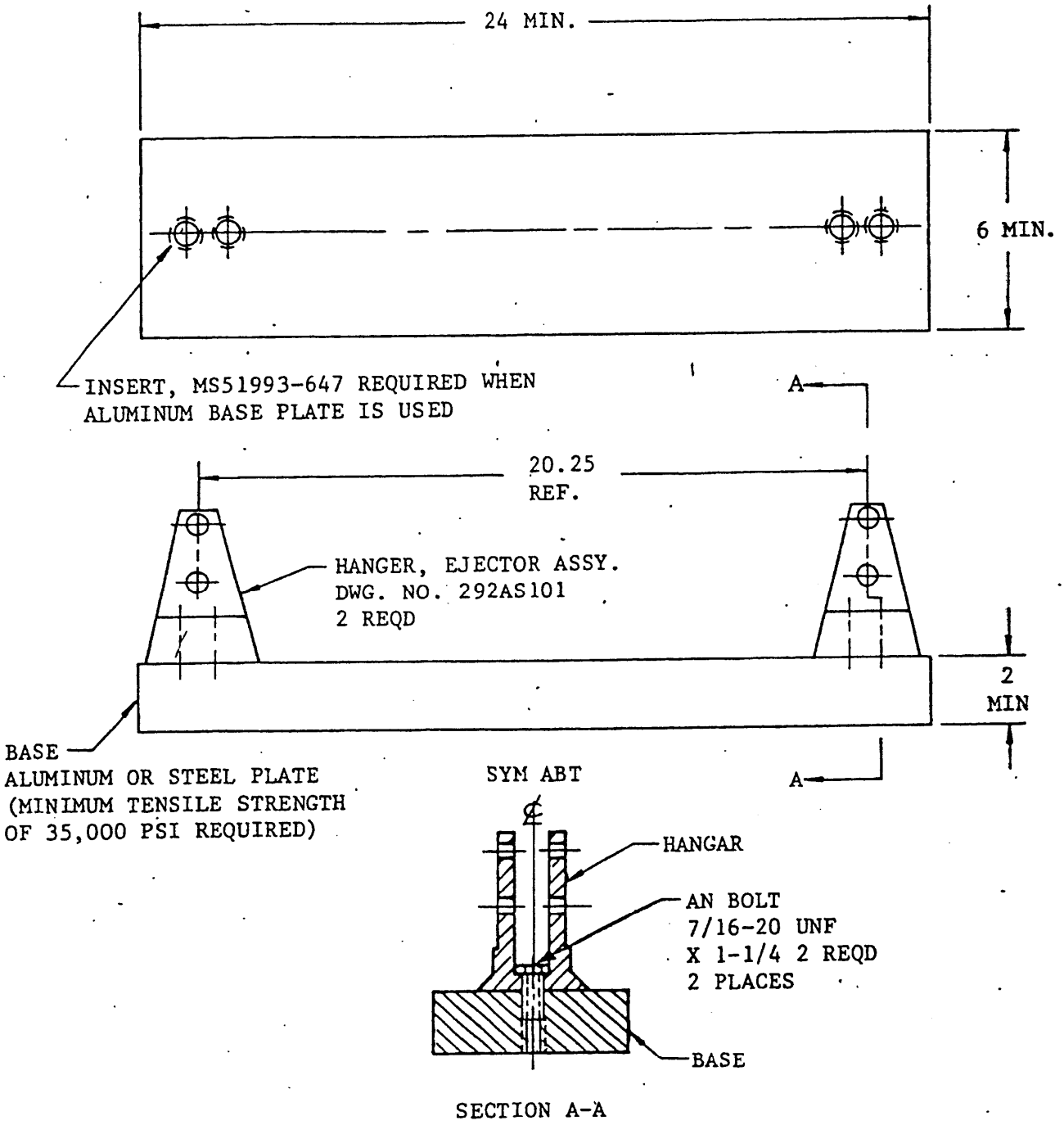
NOTE

Whenever a store is loaded onto any station of the bomb rack, the piston of each loaded ejector shall be pulled down so that it contacts the test store.

e. Test fixtures. During any test requiring that the bomb rack be supported, a fixture shall be utilized which is in accordance with DL1025AS100. During any test requiring that an individual ejector unit be supported, a fixture manufactured in accordance with Figure 1 shall be used. The hangar assemblies (Drawing 292AS101) shall be furnished by the procuring activity (see 6.2.lm).

f. Input power requirements. The bomb rack shall perform satisfactorily under all applicable conditions specified herein from a 28 VDC rated voltage

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NOTE: DIMENSIONS ARE IN INCHES

FIGURE 1. Ejector unit mounting fixture.

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power source with characteristics as defined in MIL-STD-704. The bomb rack shall operate over a range of 21 to 29 VDC, measured at the bomb rack's input connector(s).

4.6 Inspection.

4.6.1 Bomb rack inspection. The bomb rack shall be inspected as follows:

4.6.1.1 Hooks. Prior to installation in the bomb rack each hook shall be subjected to the following inspections:

a. Each hook shall be thoroughly examined to determine conformance to all the requirements of the drawings listed in the data list for the bomb rack being procured (see 6.2.lf). Any of the following major defects shall be cause for rejection.

1. One or more dimensions out of tolerance.

2. Surface texture not in accordance with specified values.

b. Each hook shall be subjected to dye penetrant inspections in accordance with MIL-I-6866. Dye penetrant inspection shall be performed after final processing. Special attention shall be given to the throat and pivot hole areas of the hooks. Evidence of cracks, seams, discontinuity of surface or other detrimental defects in any area of the hook shall be cause for rejection.

4.6.1.2 Bomb racks. Each bomb rack shall be inspected to ensure that it has been properly assembled and adjusted, that the electrical connections are secure and that the workmanship is as specified herein (see 3.7).

4.6.1.3 Qualification by analysis, inspection, and demonstration. All requirements for the bomb rack not qualified by testing shall be qualified by analysis, inspection, or demonstrations approved by the procuring activity (see 3.6.2a).

4.6.1.4 Packaging inspection. Each bomb rack shall be examined to ensure that the preparation for delivery conforms to this specification (see section 5).

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4.6.2 Functional test. Each bomb rack or ejector unit shall pass the following tests. Mounting of ejector units not attached to the Type I or Type II strongback shall be the same as if attached to a strongback centerline station.

WARNING

The bomb rack shall not be fired without a reactive load imposed on the ejector foot.

4.6.2.1 Ejector unit. Prior to assembly to the main Type I or Type II strongback, each ejector unit shall be functionally tested as follows. During first article and quality conformance inspections the ejector units may be removed from the Type I or Type II strongback for the purpose of conducting the test.

a. The hooks of the ejector unit shall be closed and it shall be verified that the hooks latch.

b. Rotate the safety stop lever to “locked”. Attempt to release the ejector unit using the manual release lever. The maximum force applied shall be 200 Newtons (45 pounds) in the forward direction. The release mechanism shall not actuate.

c. Rotate the safety stop lever to “unlocked”. Release the ejector unit using the manual release lever. The release force shall be between 67 and 200 Newtons (15 and 45 pounds). The release mechanism shall operate smoothly and the hooks shall open.

d. The hooks of the ejector unit shall be latched and a total down load of 22,240 Newtons (5,000 pounds) (11,120 Newtons, 2,500 pounds per hook) shall be applied to the ejector unit. There shall be no damage to the ejector unit. Rotate the safety stop lever to the “locked” then “unlocked” position. There shall be no interference or binding.

e. The load on the ejector unit shall be reduced to 3,558 Newtons (800 pounds) (1,779 Newtons, 400 pounds per hook). Release the ejector unit using the manual release lever. The release force shall be between 67 and 200 Newtons (15 and 45 pounds). The release mechanism shall operate smoothly and the hooks shall open.

4.6.2.2 Bomb rack, mechanical function. Each complete bomb rack shall be functionally tested as follows:

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a. Suspend a 113 kilogram (250 pound) store from each ejector unit, rotate the safety stop lever to "locked," and adjust the swaybraces on each ejector unit as follows:

1. Hand tighten all braces until the pads lightly contact the store.

2. Tighten the forward starboard pad 1/4 turn.

3. Tighten the aft port pad 1/4 turn.

4. Tighten the forward port pad 1/4 turn.

5. Tighten the aft starboard pad 1/4 turn.

6. Repeat tightening in order of 2. through 5. to attain 3/8 total turn on each swaybrace.

b. Insert the electrical safety pin. Set the stepping switch in the ripple mode and load each ejector unit with the applicable impulse cartridge (see 6.2.lk). Rotate the safety stop lever on each ejector unit to “unlocked." Apply 21 VDC to the bomb rack. The cartridges shall not ignite.

c. Remove the electrical safety pin. Apply 21 VDC to the bomb rack. The impulse cartridges shall ignite, and the stores shall be ejected in the sequence of 4.6.2.3c4. The government approved testing laboratory shall verify that the time of release or lag time, as defined in MIL-T-7743, conforms to the following:

1. Type I bomb rack - 0.3 second maximum.

2. Type II bomb rack - 0.6 second maximum.

3. Each individual ejector unit - 0.075 second maximum.

d. Remove, disassemble, clean and inspect the ejector assembly and surrounding areas of the bomb rack as indicated in the applicable manuals (see 6.2.lg). There shall be no damage to the bomb rack.

4.6.2.3 Bomb rack, electrical function. The bomb rack shall be attached to the vibration exciter table by means of a test fixture (see 4.5e). The vibration shall be applied in the transverse axis only. The

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input acceleration level shall be 2g's ±10 percent at a frequency of 50 ±10 Hz. Vibration parameter tolerances shall be in accordance with MIL-T-7743. Acceleration levels shall be measured at locations adjacent to the bomb rack mounting points on the test fixture. Care shall be exercised to minimize resonances produced by basic bomb rack structural components. Resonances produced by secondary structural items (brackets, clips, etc.) are acceptable. The total time of vibration shall not exceed 5 minutes. During vibration, the following electrical tests shall be completed. Interruption of the vibration is acceptable, if required, to allow time for operation of switches, etc. The nomenclature used to describe the electrical components are in accordance with the applicable schematic diagrams (see 6.2.11). Unless otherwise stated the test voltage shall be 21 -0 VDC.

a. Check for stepper switch homing circuit. The forward hooks shall be closed on all ejector units.

1. Apply the test voltage to pin 4 (positive) and pin 7 (ground) of connector J1.

2. Hold the step home switch in the “step” position to move the stepper switch out of the home position.

3. Release the step home switch (off position). Depress the homing light and the bulb shall illuminate.

4. Hold the step home switch in the homing position. After the stepper switch reaches the home position, the homing light shall illuminate.

5. Release the step home switch (off position). The homing light shall not illuminate.

6. Remove the voltage from connector J1.

b. Check for automatic homing circuit. The forward hooks shall be closed on all ejector units.

1. Apply the test voltage to pin 4 and pin 7 of connector J1. The stepper switch shall automatically home (a momentary pulse is applied to the homing light).

2. Apply the test voltage to pin 8 (positive) of connector J1. Voltage shall exist between the center contact and ground of the J7 connector

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(Type I bomb rack) or the J13 connector (Type II bomb rack).

3. Remove the voltage from pin 8 of connector J1.

4. Remove the test voltage from pin 4 and pin 7 of connector J1 and then reapply. The stepper switch shall automatically home to station number 1 (centerline ejector unit on the Type I bomb rack, aft centerline ejector unit on the Type II bomb rack).

5. Repeat steps 2, 3, and 4 two additional times.

6. Remove all voltages from connector J1.

c. Check the store release circuits. The forward hooks shall be closed on all ejector units.

1. Apply the test voltage to pin 4 (positive) and pin 7 (ground) of connector J1.

2. Apply the test voltage to pin 8 of connector J1. Voltage shall exist between the center contact and ground of the J7 connector (Type I bomb rack) or the J13 connector (Type II bomb rack).

3. Remove the test voltage from pin 8 of connector J1.

4. Repeat this procedure to check the proper sequencing of the test voltage to all ejector units. The firing order shall be as follows:

Type I Bomb Rack	Type II Bomb Rack
Center station (J7 connector)	Center aft station (J13 connector)
Left station (J6 connector)	Center forward station (J12 connector)
Right station (J5 connector)	Left aft station (Jll connector)
	Left forward station (J10 connector)
	Right aft station (J9 connector)
	Right forward station (J8 connector)

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The stepper switch shall automatically home to station 1 after application of the firing pulses.

5. Repeat steps 1, 2, 3 and 4 two additional times.

6. Remove all voltages from connector J1.

d. Check the rocket and CBU firing circuits. The forward hooks shall be closed on all ejector units.

1. Apply the test voltage to pin 4 and pin 7 of connector J1.

2. Apply the test voltage to pin 6 (positive) of connector J1 (the select rockets relay shall be energized).

3. Apply the test voltage to pin 8 of connector J1. Voltage shall exist between pins A and E of connector J4 (Type I bomb rack) or connector J7 (Type II bomb rack).

4. Remove the voltage from pin 8 of connector J1.

5. Repeat steps 3 and 4 to check the proper sequencing of the test voltage to all rocket and CBU firing circuits. The order shall be as follows:

Type I Bomb Rack	Type II Bomb Rack
Center station (J4 connector)	Center aft station (J7 connector)
Left station (J3 connector)	Center forward station (J6 connector)
Right station (J2 connector)	Left aft station (J5 connector)
	Left forward station (J4 connector)
	Right aft station (J3 connector)
	Right forward station (J2 connector)

6. Remove all voltages from connector J1.

7. Repeat steps 1 through 6 two additional times.

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e. Check the store automatic step circuits. The hooks of the center ejector unit (Type I bomb rack) or center aft ejector unit (Type II bomb rack) shall be closed. The hooks of the remaining ejector units shall be open.

1. Apply the test voltage to pin 4 and pin 7 of connector J1.

2. Apply the test voltage to pin 8 of connector J1. Voltage shall exist between the center contact and ground of the J7 connector (Type I bomb rack) or the J13 connector (Type II bomb rack).

3. Remove the test voltage from pin 8 of connector J1.

4. Repeat steps 2 and 3.

5. Remove all voltages from connector Jl, and open the hooks of the center ejector unit (Type I bomb rack) or center aft ejector unit (Type II bomb rack).

6. Close the hooks of the next ejector unit in the firing sequence (see c.4). Repeat steps 1 through 5.

7. Repeat step 6 until all stations have been checked.

f. Check the CBU sensing circuit. The hooks of all the ejector units shall be closed.

1. Place a 10 ohm, 100 watt resistor across pins A and E of connector J4 (Type I bomb rack) or connector J7 (Type II bomb rack).

2. Apply the test voltage to pin 4 and pin 7 of connector J1.

3. Apply the test voltage to pin 6 of connector J1 (the select rockets relay shall be energized).

4. Position the mode selector switch in the CBU position. Apply the test voltage to pin 8 of connector J1. The voltage acres-s the resistor shall be between 18 and 21 volts.

5. Remove all voltages from connector J1.

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g. Check the over-ride relay circuit. The forward hooks of all the ejector units shall be closed.

1. Place a 10 ohm, 100 watt resistor across pins A and E of connector J4 (Type I bomb rack) or connector J7 (Type II bomb rack).

2. Place a jumper wire across pins C and E on one of the forward station connectors for the Type I bomb rack (J2 or J3) or one of the remaining station connectors for the Type II bomb rack (J2, J4 or J6).

3. Apply the test voltage to pin 4 and pin 7 of connector J1.

4. Apply the test voltage to pin 6 of connector J1 (the select rocket relay shall be energized).

5. Position the mode selector switch in the CBU position.

6. Apply the test voltage to pin 8 of connector J1. The voltage across the resistor shall be between 11 and 16 volts.

7. Position the mode selector switch to the rocket normal position. The voltage across the resistor shall remain between 11 and 16 volts.

8. Remove all voltages from connector J1.

h. Check the ejector unit transfer circuit. The forward hooks of all the ejector units shall be closed.

1. Place a 28 VDC lamp between pin 4 and pin 5 of connector J1.

2. Apply the test voltage to pin 4 and pin 7 of connector J1. The lamp shall illuminate.

3. Apply three firing pulses to the Type I bomb rack, or six firing pulses to the Type II bomb rack (application and removal of test voltage on pin 8 of connector J1). The lamp shall remain illuminated for the firing pulses and momentarily go out upon removal of the last firing pulse.

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4. Repeat steps 1 through 3 two additional times.

5. Remove all voltages from connector J1.

i. Check the safety pin circuit.

1. Measure the resistance between pin 8 and pin 4 of connector J1. The electrical safety pin shall not be installed. The resistance shall be between 500 and 700 ohms.

2. Measure the resistance between pin 8 and pin 4 of connector J1 with the electrical safety pin installed. The circuit shall indicate open.

j. Check the reset circuit. The forward hooks of all the ejector units shall be closed.

1. Apply the test voltage to pin 4 and pin 7 of connector J1.

2. Apply a test voltage pulse to pin 8 of connector J1 (the stepper switch will no longer be in the "home" position).

3. Apply the test voltage to pin 11 of connector J1. Remove the voltage from pin 11 and the stepper switch shall home.

4. Apply the test voltage to pin 8 of connector J1. Voltage shall exist between the center contact and ground of the J7 connector (Type I bomb rack) or the J13 connector (Type II bomb rack).

5. Remove all voltages from connector J1.

k. Check the electromechanical arming circuits.

1. Apply the test voltage to pin 2 (positive) and pin 7 of connector J1. Verify that all the forward electromechanical arming devices of each ejector unit have been energized (actuation of the device's locking system).

2. Remove the test voltage from pin 2 and pin 7 of connector J1.

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3. Apply the test voltage to pin 3 (positive) and pin 7 of connector J1. Verify that all the aft electromechanical arming devices of each ejector unit have been energized (actuation of the device's locking system).

4. Remove the test voltage from pin 3 and pin 7 of connector J1.

l. Remove the bomb rack from the vibration exciter table. The remaining tests shall be completed under static conditions. Unless otherwise specified, the remaining tests shall be conducted with the selector switch in the rocket normal mode.

m. Check the firing diode. The hooks of all the ejector units shall be closed.

CAUTION

With the firing power polarity reversed, care should be exercised against short circuits that may exist due to common ground between the power supply and equipment.

1. Apply the test voltage so that pin 4 and pin 8 are negative and pin 7 is positive (connector Jl). Voltage shall not exist between the center conductor and ground for connector J7 (Type I bomb rack) or connector J13 (Type II bomb rack).

2. Remove all Voltages from connector J1.

n. Check the electric fuzing system of each ejector unit. The hooks of all the ejector units shall be open.

1. Apply the test voltage to the electric fuze input connector J25.

2. Place a 28 VDC lamp between the center and outer conductors of the MK-39 receptacle of each ejector unit (J26 connector of each station). The lamp shall illuminate.

3. Close the forward hook of each ejector unit. The lamp shall go out.

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4. Open the forward ejector unit and place a 5.55 $^{+0.794}_{-0}$ (7/32 $^{-0}_{}$ inch) spacer between the hook toe and the ejector housing as shown in Figure 2. The lamp shall illuminate at this hook position.

o. Check the auto ripple release mode.

1. Apply the test voltage to pin 4 and pin 7 of connector J1.

2. Hold the step-home switch in the homing position. The stepper switch shall go to the home position and the homing light shall illuminate.

3. Place the selector switch to the auto ripple mode.

4. Apply the test voltage to pin 8 of connector J1. The stepper switch shall continuously operate, providing firing pulses to each ejector unit station of the bomb rack. The stepper switch shall continue to operate with the test voltage applied to pin 8.

5. Remove all voltages from connector J1.

p. Check the mechanical step release mode.

1. Apply the test voltage to pin 4 and pin 7 of connector J1.

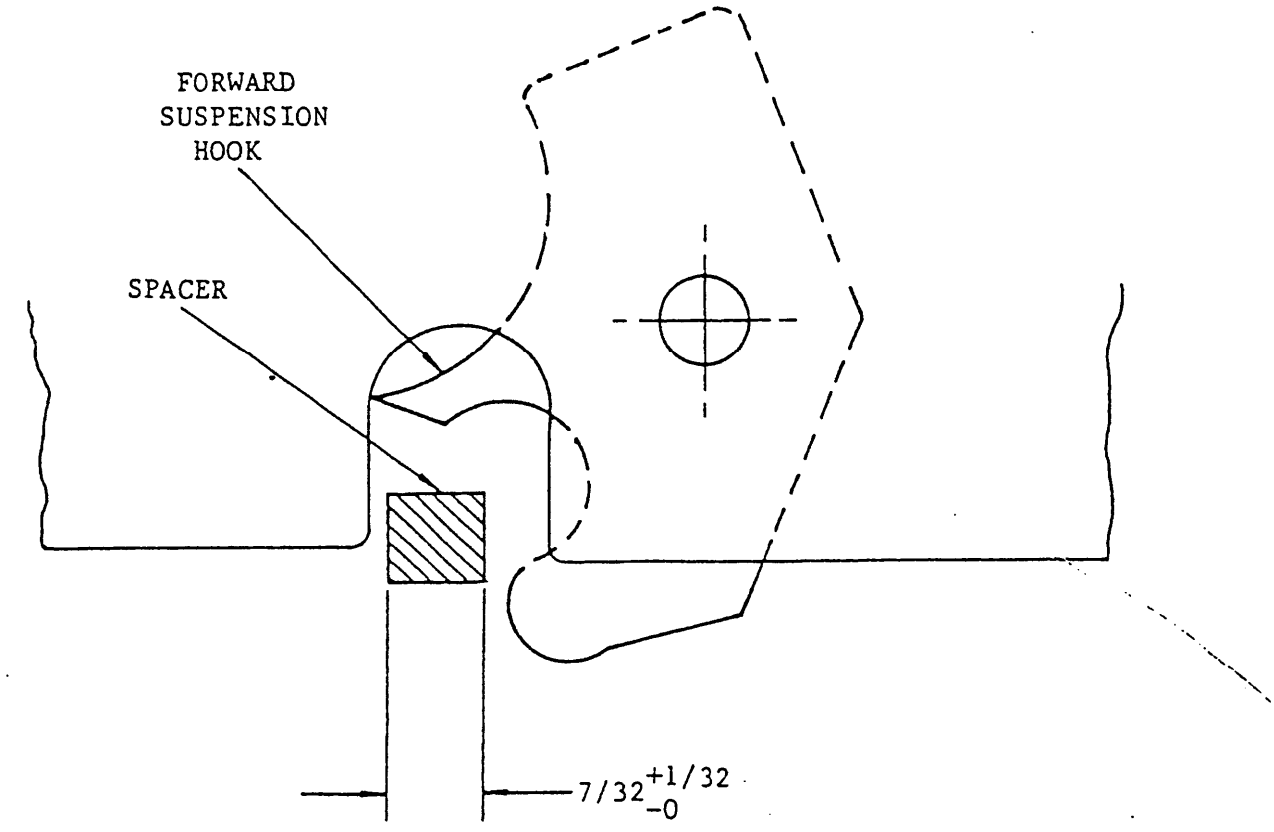
2. Hold the step-home switch in the homing position. The stepper switch shall go to the home position and the homing light shall illuminate.

3. Momentarily apply the test voltage to pin 9 (positive) of connector J1.

4. The stepper switch shall step one station for each application of the test voltage to pin 9 of the J1 connector. Check to ensure that voltage does not exist at the station where the stepper switch is located.

5. Remove all voltages from connector J1.

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NOTE: DIMENSIONS ARE IN INCHES.

FIGURE 2. Spacer location.

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q. Check the select rocket return circuit.

1. Place a 28 VDC lamp between pin 10 (positive) and pin 7 of connector J1.

2. Apply test voltage to pin 6 and pin 7 of connector J1. The lamp shall illuminate.

3. Remove all voltages from connector J1.

r. Perform a dielectric test on all circuits of the bomb rack.

1. Apply 1,000 ±50 volts, 60 Hz (rms) for 1 minute between all combinations of pins of connector J1. Any arc-over, breakdown or leakage current in excess of 1 milliampere shall constitute a failure.

2. Apply 1,000 ±50 volts, 60 Hz (rms) for 1 minute between all unconnected combinations of pins of each rocket and CBU harness. Any arc-over, breakdown or leakage current in excess of 1 milliampere shall constitute a failure.

4.6.3 Arming mechanism. A MK-9 or MK-11 bomb arming wire assembly shall be inserted in the tail and nose arming units of each ejector unit (see 6.2.1m). With each arming unit not energized, pull each arming wire slowly, in a vertically downward direction, parallel to the side of the ejector unit. Each arming unit shall support a 40 Newton. (9 pound) load, but release the wire at any load in excess of 58 Newtons (13 pounds). A minimum of three successive measurements shall be obtained on each non-energized arming unit. With each arming unit energized, pull on each arming wire in the same direction as specified above. Each arming unit shall support a load of 667 Newtons (150 pounds).

4.6.4 Electric fuze arming. The 0° pull-out force for the electric fuze connector assembly and the pull test fixture shall be in accordance with MIL-C-81842. Three successive measurements shall be made. In addition, the interlock switch shall be verified to be open when the forward hook is latched and closed when the forward hook is open.

4.6.5 Life test. This test shall be conducted as follows:

4.6.5.1 Life test, electrical. Each ejector unit shall be monitored for proper electrical performance while the bomb rack stepping switch is cycled electrically to simulate 250 firings in the ripple mode. The test shall be conducted with the hooks closed and 28 VDC

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applied. After each 50 simulated firings the electrical safety pin shall be inserted and 28 VDC applied. The bomb rack shall not operate.

4.6.5.2 Life test, ejection. Ejector units from the bomb racks, in the quantities listed below, shall be subjected to a 250 ejection life test.

a. Type I bomb rack - 2 ejector units

b. Type II bomb rack - 3 ejector units

Ejector units may be removed from the strongback to conduct these tests. Each ejection shall be performed with a vertical down load of 22,240 Newtons (5,000 pounds) per hook (44,480 Newtons, 10,000 pounds total per ejection unit) and the applicable impulse cartridge (see 6.2.1k). The ejector foot shall be opposed by a device which simulates the force-time characteristics of an ejected MK 83 L.D.G.P. store (450 kilograms, 1,000 pounds). Time of release or lag times shall be measured as specified in 4.6.2.2c on the following ejections.

c. Ejections 1 through 10

d. Ejections 126 through 135

e. Ejections 241 through 250

4.6.6 Additional tests. The bomb rack shall pass the following tests of MIL-T-7743 as specified, except as modified herein. Applicable impulse cartridges shall be used when required (see 6.2.1k). When arming units or electric fuzes are required to be tested, only the tests of 4.6.3 or 4.6.4 respectively shall apply.

a. Electrical Characteristics (see 4.6.6.1)

b. Lag Time or Time of Release Determination (see 4.6.6.2)

c. Icing (see 4.6.6.3)

d. Temperature Shock (see 4.6.6.4)

e. Altitude (see 4.6.6.5)

f. Humidity

g. Sand Test (see 4.6.6.6)

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h. Salt Spray (see 4.6.6.7)

i. Shock (see 4.6.6.8)

j. Vibration (see 4.6.6.9)

k. Lock Shut Firing (see 4.6.6.10)

l. Static Test (see 4.6.6.11)

4.6.6.1 Electrical characteristics. The bomb rack shall be subjected to the electrical characteristics test with the following exceptions:

a. The minimum operating voltage shall be determined by demonstrating that all electrical equipment on the bomb rack operates when 21 VDC is applied to the terminals of the applicable bomb rack connectors.

b. The test temperature shall be 70°C (158°F).

4.6.6.2 Lag time or time of release determination. The bomb rack shall be subjected to this test with the following exceptions:

a. Only horizontal releases shall be conducted.

b. The stepping switch shall be positioned in the ripple mode.

c. All ejections shall be performed with a 22.7 kilogram (50 pound) simulated store suspended from each ejector unit and. swaybraced in accordance with 4.5c.

d. The high temperature test temperature shall be 70°C (158°F).

e. The firing voltage shall be 21 VDC.

After completion of the above at ambient room conditions, the bomb rack shall be subjected to the electrical function tests (see 4.6.2.3).

4.6.6.3 Icing. The bomb rack shall be subjected to the first and second freeze of the Navy/Air Force icing test. This shall be conducted on two ejector units only, removed at random from the strongback of the bomb rack. The ejector unit shall be supported as described in 4.6.2.

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4.6.6.4 Temperature shock. Stores shall be ejected using the ripple mode. The test temperature shall be 70°C (158°F).

4.6.6.5 Altitude. Stores shall be ejected using the ripple mode.

4.6.6.6 Sand test. This test shall be conducted on two ejector units only, removed at random from the strongback of the bomb rack. The ejector units shall be supported as described in 4.6.2.

4.6.6.7 Salt spray. Stores shall be ejected using the ripple mode. The minimum operating voltage shall be as defined herein (see 4.6.6.1). When pulsed power inputs are required, the on-time shall be 0.025 second and the off-time shall be 0.020 second.

4.6.6.8 Shock test. The applied shock loads shall be as specified in Table IV. There shall be two shocks in each direction of the three axes shown in the table for a total of 12 shocks. Each ejector unit shall be loaded with a 113 kilogram (250 pound) store for the yield load tests. After the yield load test in each direction of an axis, stores shall be manually released to assure proper functioning of the ejector units. The light load test shall not be conducted. Only yield loads shall be applied during catapult and arrested landing tests.

4.6.6.9 Vibration. The bomb rack shall be subjected to the vibration test in accordance with Procedure II. Stores shall not be released during vibration and 113 kilogram (250 pound) stores shall be used for loaded vibration.

4.6.6.10 Lock shut firing. This test shall be conducted on two ejector units only, removed at random from the strongback of the bomb rack. The ejector units shall be supported as described in 4.6.2.

4.6.6.11 Static test. This test shall be conducted with the following conditions. The design limit load shall be as specified in Table V. The bomb rack shall be loaded to limit, yield and ultimate loads. The combined loads and moments shall be applied to rigid simulated stores of negligible weight (45 kilograms, 100 pounds maximum), such that they act at each store center of gravity.

4.6.7 Qualification cross reference. Table VI identifies each requirement in Section 3 and correlates the Section 4 paragraphs that qualify it.

5. PACKAGING

5.1 Bomb rack preservation and packaging. Preservation and packaging shall be in accordance with Level A, B or C as specified in MIL-STD-794, as specified in the contract, without the use of preservation compound.

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5.2 Packing. Shipping containers shall contain one each packaged bomb rack. The container shall be suitable for Level A, B or C as specified in MIL-STD-794,

5.3 Unitized loading. Unitized loading for the bomb racks is authorized as specified in MIL-STD-794.

5.4 Marking. All shipping containers shall be marked in accordance with the requirements of MIL-STD-129 with the following information:

RACK, BOMB, MULTIPLE (OR TRIPLE) EJECTOR
MILITARY SPECIFICATION NO. MIL-R-85323(AS)
PRESERVED (DATE)
DOMESTIC OR EXPORT PACKED (AS APPLICABLE)
GOVERNMENT ORDER NO. (OR CONTRACT NO. IF ORDER NO. IS NOT ASSIGNED)
NAME OF SUPPLIER (AND NAME OF MANUFACTURER IF NOT THE SAME)
MANUFACTURER's CODE IDENTIFICATION NUMBER

6. NOTES.

6.1 Intended use. The bomb racks covered by this specification are intended for use in all types of bombing. They are designed to be mounted externally as on an aircraft wing or fuselage.

6.2 Ordering data. Procurement documents shall specify the following.

6.2.1 Procurement requirements.

a. Title, number and date of this specification.

b. Selection of applicable levels of preservation, packaging and packing required (see 5.1 and 5.2).

c. Samples subjected to sampling inspection shall not be considered or accepted as part of the contract [see 4.4.2).

d. Sampling inspection selection (see 4.4.2).

e. Number of samples and inspection sequence for production lots in excess of 300 units (see Table III).

f. Applicable drawing package for type of bomb rack required (see 3.5 and 3.6.1).

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g. Applicable maintenance manuals for type of bomb rack required (see 4.6.2.2).

h. Name and location of government representative responsible for random selection of inspection samples (see 4.4.2).

i. Items of data required (see 6.3).

j. Name and location of government approved test laboratory.

k. Number and designation of impulse cartridges required for testing (see 4.6.2.2, 4.6.5.2 and 4.6.6).

l. Applicable schematic diagram for type bomb rack required (see 4.6.2.3).

m. Name and location of government activities responsible for supplying hanger assemblies and arming wire assemblies (see 4.5e).

6.2.2 Contract provision. Contracts shall specify the following provision for first article inspection.

6.2.2.1 First article. When a first article is required for inspection and approval (see 3.1, 4.3, 6.2 and 6.3), the contract shall specify the following provision for first article inspection. When a contractor is in continuous production of bomb racks from contract to contract, consideration should be given to waive the first article inspections. If inspection is required, indicate:

a. If first article inspections are conducted at the contractor's plant or a government approved laboratory, an inspection report shall be forwarded to the procuring activity for verification.

b. That the approval of first article samples or the waiving of the first article inspection shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

6.3 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423)

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incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.3	First Article Inspection Reports	DI-T-5329 - Inspection Test Reports
4.4	Quality Conformance Inspection Reports	DI-T-5329 - inspection Test Reports

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.4 Quality assurance definitions. Definitions for quality assurance terms are in accordance with MIL-STD-109.

6.5 International standardization agreement. Certain provisions (3.6.1) of this specification are the subject of international standardization agreements STANAG 3575, AIR STD 20/10 and AIR STD 20/17. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Preparing Activity
NAVY-AS
DOD Proj. No. 1095-N114

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TABLE I. First article inspections - schedule and sequence. 1/

Tests	Test paragraph	Sample A	Sample B
Bomb rack inspection	4.6.1	1	1
Functional 2/	4.6.2	2	2
Arming mechanism	4.6.3	3	3
Electric fuze arming	4.6.4	4	4
Electrical characteristics	4.6.6a	5	x
Lag time or time of release determination	4.6.6b	x	5
Icing	4.6.6c	x	6
Temperature shock	4.6.6d	x	7
Altitude	4.6.6e	x	8
Humidity	4.6.6f	x	9
Sand	4.6.6g	x	10
Life	4.6.5	x	11
Salt spray	4.6.6h	x	12
Shock	4.6.6i	6	x
Vibration	4.6.6j	7	x
Lock shut firing	4.6.6k	x	13
Static	4.6.61	8	x

1/ Inspection sequence is denoted by the numbers in the sample columns. An X denotes inspection not required.

2/ The vibration requirement is not applicable during this test. Electrical function tests will be performed under static conditions only.

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TABLE II. Quality conformance inspections - schedule and sequence. 1/

Tests	Test paragraph	Sample A	Sample B	Sample C	Sample D
Group 1		All samples			
Bomb rack inspection	4.6.1	1			
Functional 2/	4.6.2	2			
Arming mechanism	4.6.3	3			
Electric fuze arming	4.6.4	4			
Group 2		A	B	C	D
Electrical characteristics	4.6.6a	1	x	1	X
Lag time or time of release determination	4.6.6b	x	1	x	1
Icing	4.6.6c	x	2	x	2
Temperature shock	4.6.6d	x	3	x	3
Altitude	4.6.6e	x	4	X	4
Humidity	4.6.6f	x	5	x	5
Sand	4.6.6g	x	6	X	6
Life	4.6.5	x	7	x	7
Salt spray	4.6.6h	x	8	x	8
Shock	4.6.6i	2	x	2	x
Vibration	4.6.6j	3	x	3	x
Lock shut firing	4.6.6k	x	9	x	9
Static	4.6.6l	4	x	4	x

1/ Inspection sequence is denoted by the numbers in the sample columns. An X denotes inspection not required.

2/ The vibration requirement is applicableonly during individual inspection tests performed by the contractor.

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TABLE III. Sampling test items.

Test	Production Lot size	Number of samples
Bomb racks	100 or less	2
	101 - 200	3
	201 - 300	4
	301 or More	1/

1/ The procuring activity shall specify the number of samples, and the sequence of inspections each unit shall be subjected to for lots in excess of 300 units (see 6.2.le).

TABLE IV. Applied shock yield loads.

Direction	Applied shock load (g's) for 113 kilogram (250 pound) store
Vertical up	17.3
Vertical down	17.3
Transverse port	8.6
Transverse starboard	8.6
Longitudinal forward	10.3
Longitudinal aft	10.3

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TABLE V. Static test limit loads.

Simulated store characteristics				
Store	Lug centers	Store description	Vertical dimension, lug bearing surface of parent bomb rack to store C.G.	
			Centerboard station	Shoulder station
A	355.6 mm. (14 in.)	MK83 (454 kg., 1,000 lb. class) Store	612 mm. (24.1 in.)	315 mm. (12.4 in.)
B	355.6 mm. (14 in.)	MK117 (340 kg., 750 lb. class) Store	556 mm. (21.9 in.)	249 mm. (9.8 in.)

Combined design-limit loads and. moments about reference point (store C.G.)						
Store	P_z Vertical	P Side	P_x Longitudinal	M_{zz} Yaw moment	M_{yy} Pitch moment	M_{xx} Rol1 moment
Type I bomb rack - three stores						
A	-12,899N (-2,900 lbs.)	10,853N (2,440 lbs.)	3,412N (767 lbs.)	-3,111N.m (-27,533 inch-lbs.)	4,911N.m (43,467 inch-lbs.)	365N.m (3,233 inch-lbs)
Type II bomb rack - six stores						
B-Fwd stores	-19,483N (-4,380 lbs.)	0 0	5, 218N (1,173 lbs.)	0 0	-400N.m (3,540 inch-lbs.)	0 0
B-Aft stores	-19,394N (4,360 lbs.)	0 0	5,218N (1,173 lbs.)	0 0	-573N.m (-5,073 Inch-lbs.)	0 0

Sign convention

$+P_x$ = Longitudinal aft
$+P_Y$ = Side to port
$+P_z$ = Vertical up
$+M_{xx}$ = Roll clockwise looking forward
$+M_{yy}$ = Pitch nose up
$+M_{zz}$ = Yaw nose to port

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TABLE VI. Qualification cross reference.

Paragraph title	Requirement paragraph	Test paragraph
First article	3.1	4.3
Selection of spec's & std's	3.2	4.6.1.3
Interchangeability	3.3	4.6.1.3
Identification of parts	3.4	4.6.1.3
Materials	3.5	4.6.1.3
Construction	3.6.1	4.6.1.3
Bomb rack inspection	3.6.2a	4.6.1
Functional test	3.6.2b	4.6.2
Arming mechanism	3.6.2c	4.6.3
Electric fuze arming	3.6.2d	4.6.4
Life	3.6.2e	4.6.5
Electrical characteristics	3.6.2f	4.6.6a
Lag time or time of release Determination	3.6.2g	4.6.6b
Icing	3.6.2h	4.6.6c
Temperature shock	3.6.2i	4.6.6d
Altitude	3.6.2j	4.6.6e
Humidity	3,6.2k	4.6.6f
Sand	3.6.2l	4.6.6g
Salt spray	3.6.2m	4.6.6h
Shock	3.6.2n	4.6.6i

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TABLE VI. Qualification cross reference (continued).

Paragraph title	Requirement paragraph	Test paragraph
Vibration	3.6.2o	4.6.6j
Lock shut firing	3.6.2p	4.6.6k
Static	3.6.2q	4.6.61
Workmanship	3.7	4.6.1.2

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